Data assimilation for wildland fires

J Mandel, JD Beezley, JL Coen… - IEEE Control Systems …, 2009 - ieeexplore.ieee.org

… A **wildland fire** interacts with the atmosphere **dynamics** … Available data include fuel-distribution **maps**, gridded atmospheric … of the **model** is the probability distribution of possible **wildfire** …

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An open multi-physics framework for modelling wildland-urban interface fire evacuations

E Ronchi, SMV Gwynne, G Rein, P Intini, R Wadhwani - Safety science, 2019 - Elsevier

… **simulation** of evacuation in WUI **wildfire** incidents, including three main modelling layers: **wildfire**, … current **wildfire** risk assessment and **mapping** tools towards dynamic fire vulnerability …

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Atmospheric interactions with **wildland fire** behaviour–I. Basic surface interactions, vertical profiles and synoptic structures

BE Potter - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… **model** with coupled fire behaviour and atmospheric … This analysis included **maps** like Fig. 7 showing the tracks of … Part II of this **review** discusses plume structure and **dynamics** in …

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Evolutionary-Statistical System: A parallel method for improving **forest fire spread prediction**

G Bianchini, P Caymes-Scutari… - Journal of Computational …, 2015 - Elsevier

… the occurrence of fire ignition and **propagation**. Every year, … to a convincing fire **propagation model**, where instructors can … , and fundamentally, students can **review** the results of their …

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Evacuation decision-making and **behavior** in wildfires: Past research, current challenges and a future research agenda

E Kuligowski - Fire Safety Journal, 2021 - Elsevier

… an understanding of human **behavior** in wildfires, which is an … decision-making and **behavior** during wildland-urban … support developments of evacuation **simulation** models, and …

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Current **wildland fire** patterns and challenges in Europe: A synthesis of national perspectives

N Fernandez-Anez, A Krasovskiy… - Air, Soil and Water …, 2021 - journals.sagepub.com

… involved in **wildland fire** research and land **management**, … fire **behavior**, linkages between weather and **wildland fire** … For example, an interactive **wildland fire** hazard GIS web **map** …

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Development and structure of Prometheus: the Canadian **wildland fire** growth **simulation model**

C Tymstra, RW Bryce, BM Wotton, SW Taylor… - Natural Resources Canada …, 2010

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Forest fire in India: a **review** of the knowledge base

S Joseph, K Anitha, MSR Murthy - Journal of forest research, 2009 - Springer

… initiatives for assessing the fire **dynamics** (for details, see the web … areas on **maps**, provisions for a crisis **management** group … in the systematic **management** of **forest fire** is limited to short-…

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Forest fire risk **prediction**: A spatial deep neural network-based framework

M Naderpour, HM Rizeei, F Ramezani - Remote Sensing, 2021 - mdpi.com

… of future **forest fire** events in respect to recent fire **behavior** such as ignition, **spread**, suppression… In this study, we set up a framework in which a **forest fire** risk **map** can be produced. The …

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Three lines of defense for **wildfire** risk **management** in electric power grids: A **review**

A Arab, A Khodaei, R Eskandarpour… - IEEE …, 2021 - ieeexplore.ieee.org

… The distance of each component from the **wildfire** can be obtained by **mapping** the **wildfire spread** path onto the power grid layout. While this distance can be accurately determined, it is …

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Application of **wildfire** risk assessment results to **wildfire** response planning in the southern Sierra Nevada, California, USA

MP Thompson, P Bowden, A Brough, JH Scott… - Forests, 2016 - mdpi.com

… We **review** general principles of effective federal fire … , growth, **behavior**, and suppression **simulation** system that uses … above to produce integrated **maps** of **wildfire** hazard for the entire …

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Integrating fire **behavior** models and geospatial analysis for **wildland fire** risk assessment and fuel **management** planning

AA Ager, NM Vaillant, MA Finney - Journal of Combustion, 2011 - Wiley Online Library

… concerns quantitative assessment and **mapping** of fire risk to … canopy fuel to meet **wildland fire management** objectives. For … with an ad hoc **review** of existing fire **behavior** modeling and …

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A national approach for integrating **wildfire simulation** modeling into Wildland Urban Interface risk assessments within the United States

JR Haas, DE Calkin, MP Thompson - Landscape and Urban Planning, 2013 - Elsevier

… **wildfire** risk analysis is establishing the likelihood of **wildfire** … of **wildfire** likelihood, which when coupled with **maps** of values… spatially-explicit **wildfire spread simulation model**. We classify …

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Efficient **forest fire** occurrence **prediction** for developing countries using two weather parameters

GE Sakr, IH Elhajj, G Mitri - Engineering Applications of Artificial …, 2011 - Elsevier

… A literature **review** of **forest fire prediction** methods divided … **prediction** systems of **forest fire** occurrence involving a large number of monitored features and comprising weather **prediction** …

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A **review** on status, implications and recent trends of **forest fire management**

V Attri, R Dhiman, S Sarvade - Archives of Agriculture and …, 2020 - academia.edu

… **Forest fire spread** out in an area having combustible material in the summer season with … locations from these sites are projected on the forest cover **map** of India to select active fire …

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A machine learning-based approach for **wildfire susceptibility mapping**. The case study of the Liguria region in Italy

M Tonini, M D'Andrea, G Biondi, S Degli Esposti… - Geosciences, 2020 - mdpi.com

… a complex and non-linear **behavior** [49,50]. At present, only a few studies exist in literature on the use of this kind of approach for **wildfire susceptibility** and risk **mapping**. Moreover, a …

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A **review** of the applications of remote sensing in fire ecology

DM Szpakowski, JLR Jensen - Remote sensing, 2019 - mdpi.com

… risk **mapping**, but also includes other factors such as fuel distribution, historical **wildfire spread**, … An example of a commonly used **model** for simulating fire growth is FARSITE [93]. The …

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Forest fire induced Natech risk assessment: A **survey** of geospatial technologies

M Naderpour, HM Rizeei, N Khakzad… - Reliability Engineering & …, 2019 - Elsevier

… to assess the risk of **forest-fire**-induced Natechs. This study is a **review** of methods based on … a systematic literature **review** of the methods used for **forest fire susceptibility**, hazard, and …

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Integrating **wildfire** risk **management** and spatial planning–A historical **review** of two Australian planning systems

C Gonzalez-Mathiesen, S Ruane, A March - International Journal of …, 2021 - Elsevier

… **wildfire** risk reduction considerations in a territorial manner across the WUI. Accordingly, more integrated approaches to **wildfire management** … interactions between **wildfire** and spatial …

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Statistical **model** for forecasting monthly large **wildfire** events in western United States

HK Preisler, AL Westerling - Journal of Applied Meteorology …, 2007 - journals.ametsoc.org

… (a), (c) **Maps** of odds indicate **forecast** odds for a particular month and year … **model** for assessing the forecasting skills of fire-danger predictors and producing 1-month-ahead **wildfire**-…

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